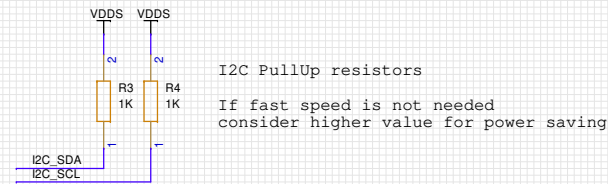
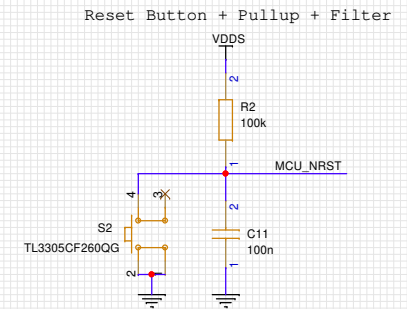
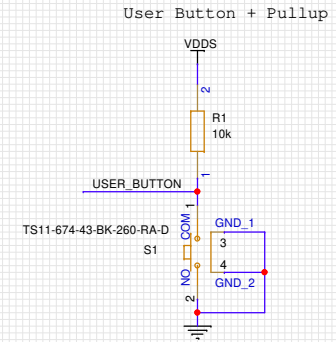
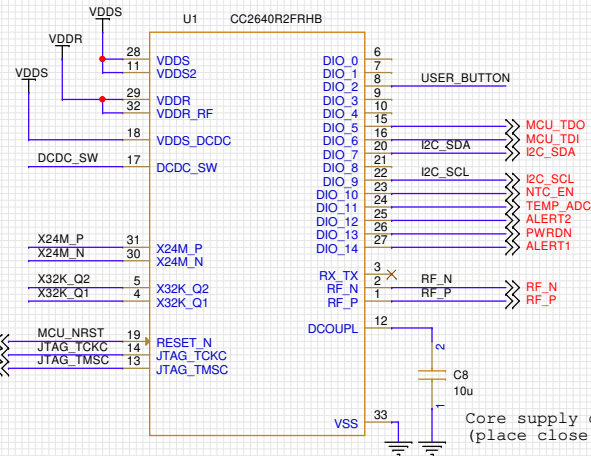
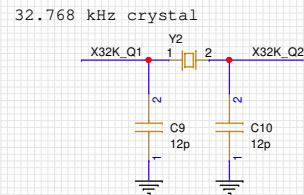
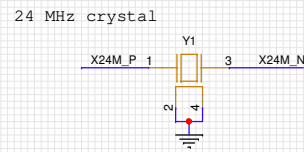
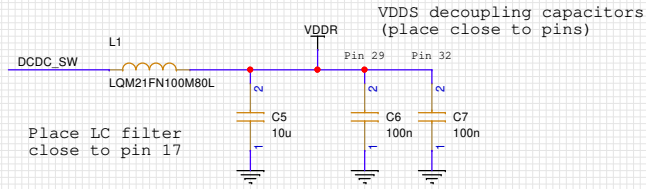
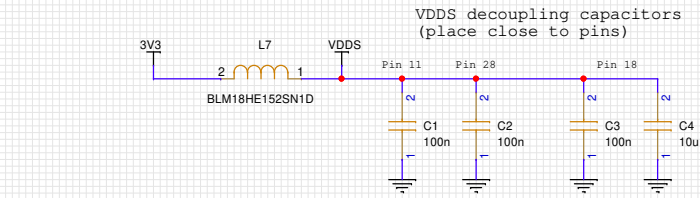


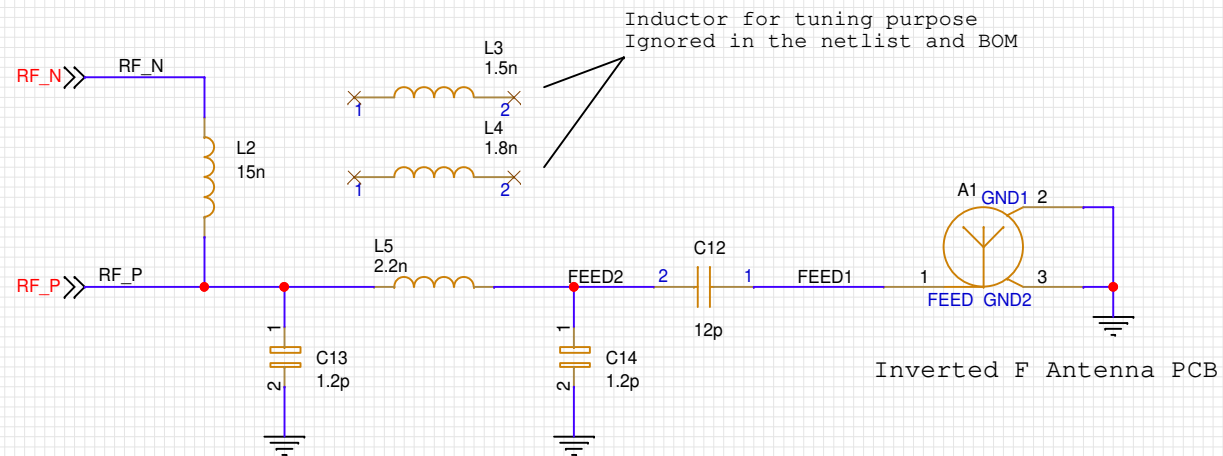
MCU



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RF Interface

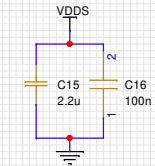
Single-Ended mode RF circuit
(reference design CC2650EM-4XS)



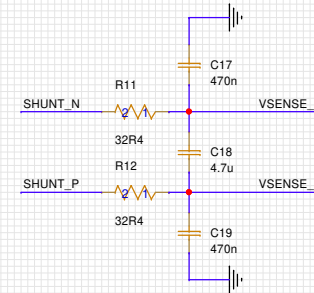
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A vertical number line with a grid. The line is labeled A at the bottom, B, C, and D at the top. A horizontal tick mark is located between B and C. An arrow points to the grid line immediately above B.

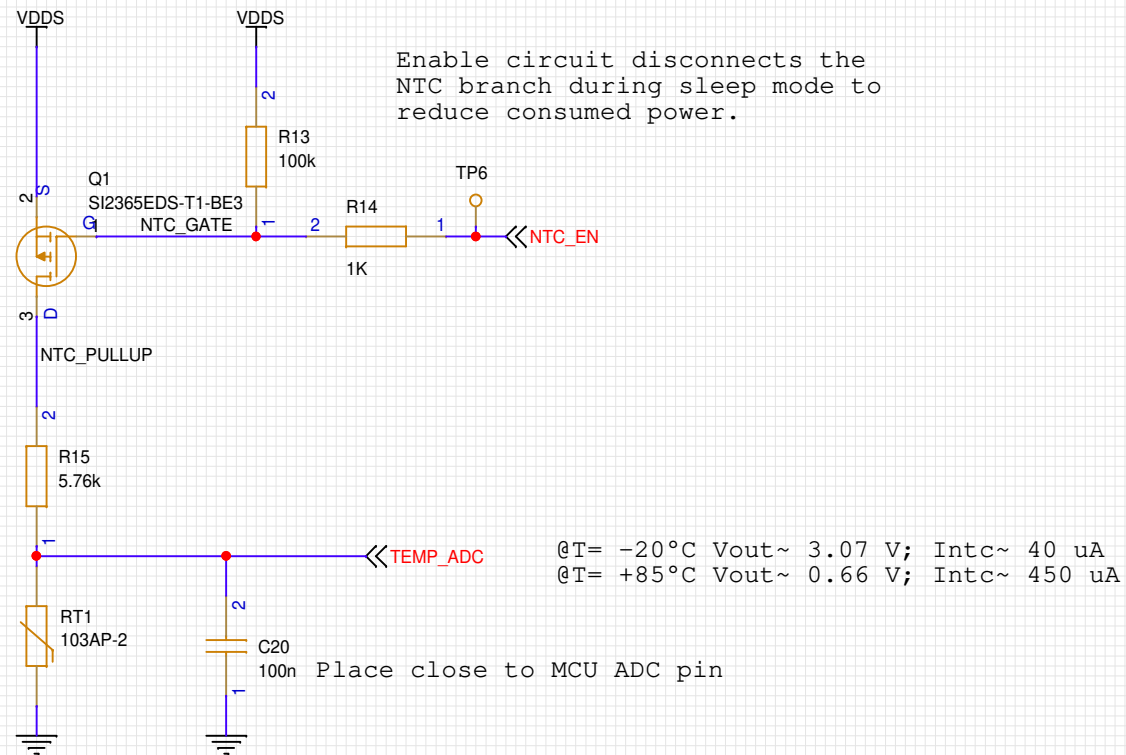
RESISTOR (1%)	SMBus Address
0 (GND)	0010 000 (R/W)
499	0010 001 (R/W)
806	0010 010 (R/W)
1,270	0010 011 (R/W)
2,050	0010 100 (R/W)
3,240	0010 101 (R/W)
5,230	0010 110 (R/W)
8,450	0010 111 (R/W)
13,300	0011 000 (R/W)
21,500	0011 001 (R/W)
34,000	0011 010 (R/W)
54,900	0011 011 (R/W)
88,700	0011 100 (R/W)
140,000	0011 101 (R/W)
226,000	0011 110 (R/W)
Tie to V _{DD}	0011 111 (R/W)



The diagram shows a shunt resistor R10, model LVK12R010CER, used for current measurement. It is connected in series between the input voltage V_{BUS_IN} and the output voltage V_{BUS_OUT} . The shunt has four pins: pin 1 is connected to the negative terminal of the shunt (SHUNT_N), and pin 4 is connected to the positive terminal (SHUNT_P). The internal resistance of the shunt is indicated as 10 mΩ.



NTC Temperature Sensor



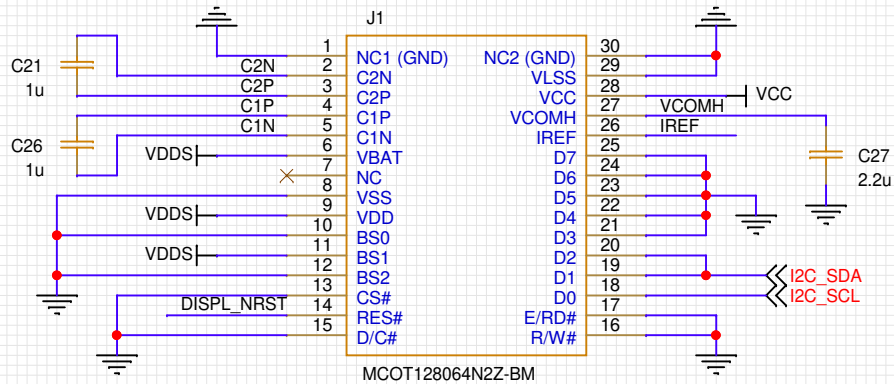
NTC: 10k @T= 25°C 0.5%
Rpullup: 5.76k <= 0.25%

To achieve <1°C @ -20°C < T < +85 °C

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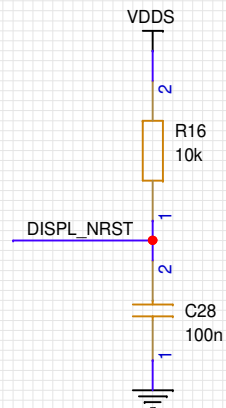
I2C OLED Display Interface

OLED Display with internal driver SSD1306



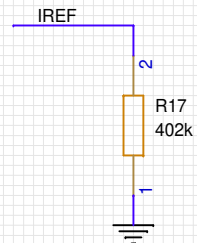
D/C#: In I2C mode, this pin acts as SA0 for slave address selection. The device will respond to the slave address following with the following byte format:

b7	b6	b5	b4	b3	b2	b1	b0
0	1	1	1	1	0	SA0	R/W#

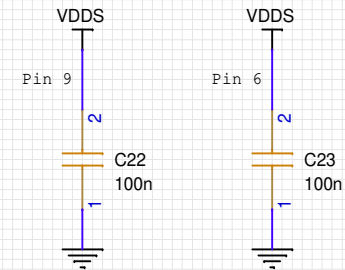


IREF regulates the segment current

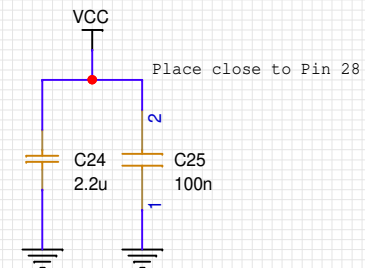
To achieve $I_{REF} = 12.5 \mu A$:
 $R = (VCC - 2.5 V) / I_{REF}$



Bypass capacitors for Display Logic and Charge Pump Input



VCC = 7.5 V for Segment Power internally generated by the SSD1306 Charge Pump circuit



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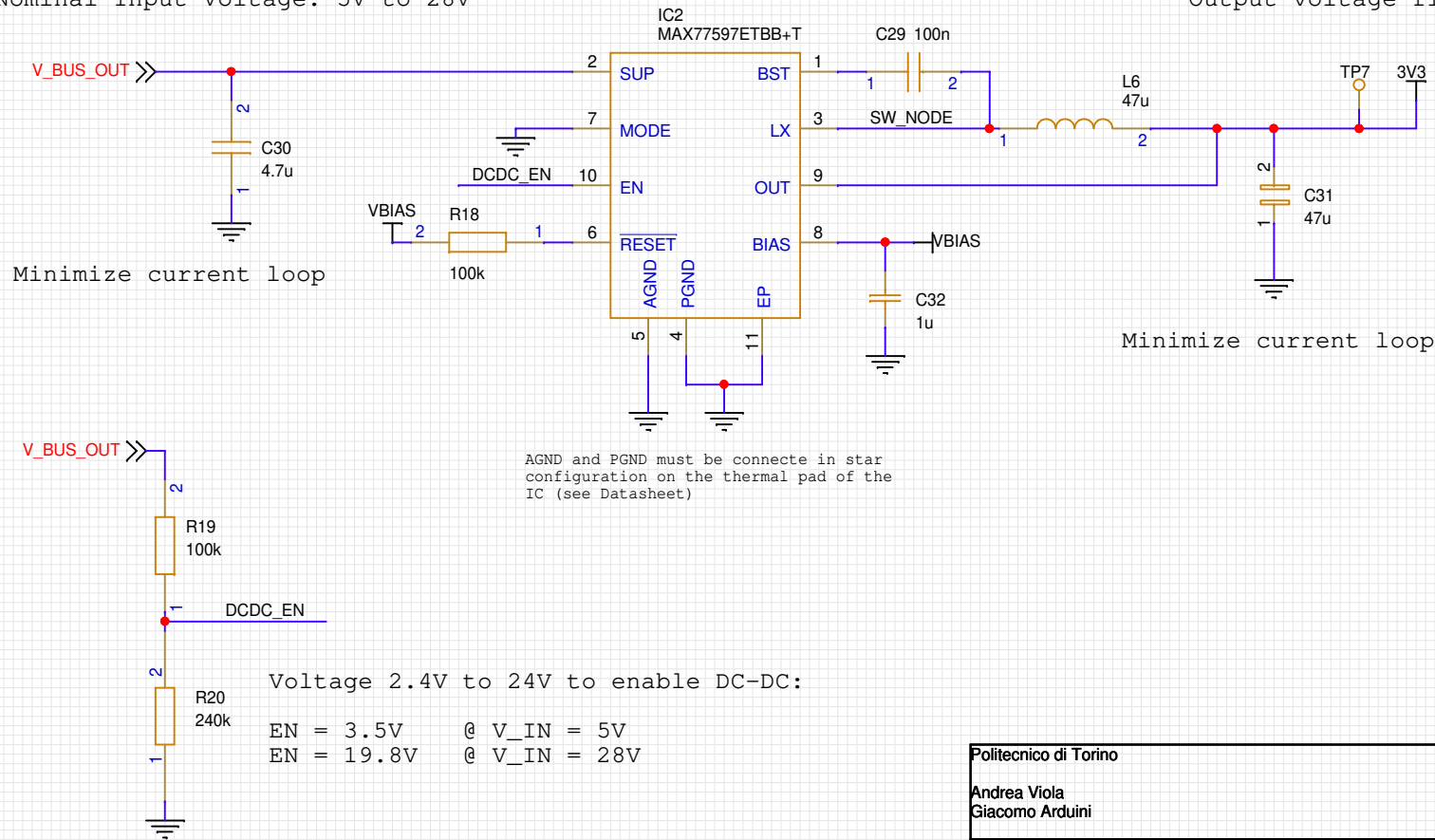
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DC-DC Buck Converter

Nominal Input Voltage: 5V to 28V

Output voltage fixed @ 3.3V



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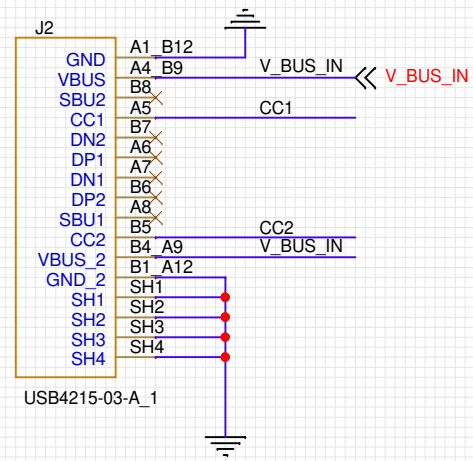
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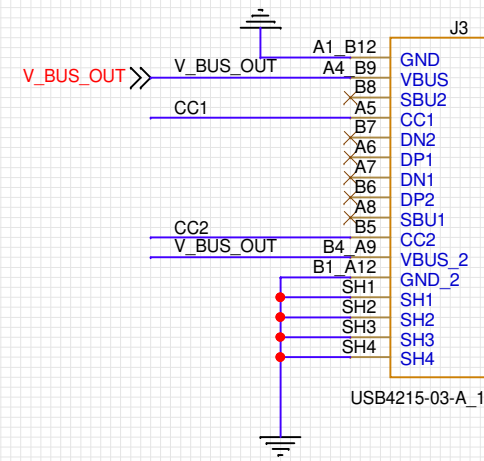
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Board Connectors

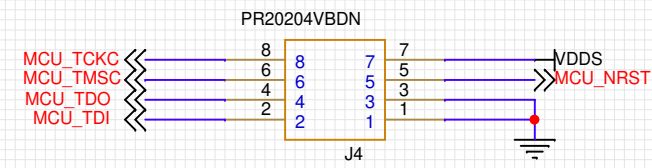
IN -> PowerBank Side



OUT -> Device Side

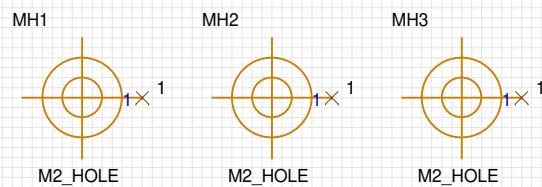


JTAG Connector Header



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Mechanical



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